


```

eta -> x1@lo1 x2:x4;      # measurement model
x3 ~~ x4;                 # correlated residual
y ~ eta m eta*m;         # structural model with latent interaction
SIMPLE: eta | m;          # simple intercepts and slopes
BURN: 10000;              # warm-up iterations
ITER: 10000;              # analysis iterations
SEED: 90291;              # random number seed

```

A brief description of several major Blimp commands follows, not all of which appear in the previous script. The software's [User Guide](#) has detailed descriptions of all Blimp commands, including many not shown here, along with dozens of analysis examples.

DATA: The DATA command specifies the input data set, which must be saved as a .csv (comma separated values) format or a whitespace (including tab) delimited file (e.g., .dat or .txt). Blimp accepts only numeric characters for data values (e.g., a nominal variable cannot have alphanumeric labels as score values), although alphanumeric characters (e.g., NA) can be used for missing value codes. Variable names can appear in the column headers, but the VARIABLES command (described next) must be omitted. No file path is needed if the Blimp script (the .imp file) is located in the same directory as the data.

VARIABLES: The VARIABLES command specifies the variable names for the data set listed on the DATA command in the input file. This command should not be used if the data file has variable names as column headers. The variable list may include variables that are not used in an analysis.

MISSING: The MISSING command is used to specify the missing value code. Missing values can be coded with a single numeric (e.g., 999) or alphanumeric value (e.g., NA).

ORDINAL: The ORDINAL command identifies binary or ordinal variables that appear in a MODEL statement. By default, Blimp adopts a probit link.

NOMINAL: The NOMINAL command specifies nominal variables that appear in a MODEL statement. Nominal variables must be represented as a single variable with numeric codes. Blimp automatically recodes the discrete responses into a set of dummy codes during estimation, where the first (lowest) code is the reference category. By default, Blimp adopts a probit or logit link function depending on whether the variable is an exogenous predictor or dependent variable.

COUNT: The COUNT command identifies count variables that appear in a MODEL statement. By default, Blimp adopts a negative binomial link function.

LATENT: The LATENT command is used to define latent variables (e.g., factors in a measurement model) that will be referenced in the MODEL section. Latent variables can be specified at any level of a multilevel model. This specification references cluster-level identifier variables from the CLUSTERID line.

CLUSTERID: The CLUSTERID command specifies cluster-level identifier variable(s) needed for a multilevel analysis or multilevel imputation. Two-level analyses require a single identifier for the

level-2 sampling unit (cluster), and three-level analyses require level-2 and level-3 identifier variables. The order of the identifier variables does not matter, as Blimp automatically determines variable levels.

CENTER: The CENTER command is used to center predictor variables in regression equations. Predictor variables in a multilevel regression model can be centered at the grand means or latent group means.

MODEL: The MODEL command typically consists of one or more regression models. Blimp’s modeling framework can accommodate a wide range of analyses ranging from basic multiple regression models to complicated multilevel structural equation models with interactions involving latent variables.

SIMPLE: The SIMPLE command is used to request conditional effects (e.g., simple intercepts and simple slopes) from a regression model with an interaction effect. When using `rbliimp`, the `simple_plot()` function can be used to graph simple intercepts and slopes from a model with an interaction, and the `jn_plot()` function produces Johnson–Neyman regions of significance. The **PARAMETERS** command can also be used to compute contrasts.

BURN: The BURN command specifies the number of burn-in or warm-up iterations. Each MCMC chain (two by default) completes the specified number of iterations before saving estimates for the final parameter summaries.

ITER: The ITER command specifies the total number of iterations for the analysis summary tables. The total number of iterations is distributed equally across the number of MCMC chains (the default is two).

SEED: The SEED command specifies the random number seed for MCMC estimation. this value must be an integer.

With the exception of variable attributes stored in the R data frame and missing data code, the commands in a Blimp Studio script correspond to input arguments in the `rblimp()` function.

```
mymodel <- rblimp(  
  data = example,  
  ordinal = 'x1:x4',  
  nominal = 'm',  
  latent = 'eta',  
  model = '  
    eta ~~ eta@1;  
    eta ~ m;  
    eta -> x1@lo1 x2:x4;  
    x3 ~~ x4;  
    y ~ eta m eta*m;',  
  simple = 'eta | m',  
  burn = 10000,  
  # rblimp function call  
  # R data frame  
  # define ordinal variables  
  # define multicategorical variables  
  # define latent variable  
  # regression models  
  # factor variance  
  # latent on predictor  
  # measurement model  
  # correlated residual  
  # model with latent interaction  
  # simple intercepts and slopes  
  # warm-up iterations
```

```
I      ter = 10000,          # analysis iterations
      seed = 90291          # random number seed
)

# print output
output(mymodel)

# plot simple slopes
simple_plot(y ~ eta | m.1 + m.2, mymodel)
```

The rblimp package has several graphing functions that are not available in Blimp Studio graphical interface. The `simple_plot()` function in the last line is one example, which graphs the simple slopes of the latent variable for each categorical group.